k-Means Clustering
– An Unsupervised Learning Algorithm

Clustering Problem:

Given a training set $\{x^{(1)}, x^{(2)}, \ldots, x^{(n)}\}$ with d features, where $x^{(i)} \in \mathbb{R}^d$ for each i,
we want to group the data into k clusters (where k is a fixed natural number).

Here we will see only the basic k-means clustering algorithm.

Indicator function:

For a statement P,

$$\mathbf{1}\{\mathrm{P}\} = \begin{cases} 1, & \text{if P is true} \\ 0, & \text{if P is false} \end{cases}$$

k-Means Clustering Algorithm:

1. Initialize cluster centroids $\mu_1, \mu_2, \ldots, \mu_k$ randomly.
2. Repeat until convergence: {

For every i $(1 \leq i \leq n)$, set

$$c^{(i)} = \operatorname{argmin}_j \left\| x^{(i)} - \mu_j \right\|^2$$

For every j $(1 \leq j \leq k)$, set

$$\mu_j = \frac{\sum_{i=1}^{n} \mathbf{1}\{c^{(i)} = j\}\, x^{(i)}}{\sum_{i=1}^{n} \mathbf{1}\{c^{(i)} = j\}}$$

}

Example:

Given the set of points $\{(1, 5), (2, 4), (4, 1), (5, 2), (2, 3), (6, 1)\}$ in $\mathbb{R}^2$, implement k-means clustering algorithm by taking k = 2 and the initial centroids as $\mu_1 = (1, 0)$, $\mu_2 = (4, 0)$.

Iteration 1:

Point	Distance to (1,0)	Distance to (4,0)	Cluster assigned
(1, 5)	$\sqrt{25}$	$\sqrt{34}$	1
(2, 4)	$\sqrt{17}$	$\sqrt{20}$	1
(4, 1)	$\sqrt{10}$	$\sqrt{1}$	2
(5, 2)	$\sqrt{20}$	$\sqrt{5}$	2
(2, 3)	$\sqrt{10}$	$\sqrt{13}$	1
(6, 1)	$\sqrt{26}$	$\sqrt{5}$	2

Updating the centroids:

$$\mu_1 = \frac{(1,5)+(2,4)+(2,3)}{3} = \left(\frac{5}{3}, 4\right)$$

$$\mu_2 = \frac{(4,1)+(5,2)+(6,1)}{3} = \left(5, \frac{4}{3}\right)$$

Iteration 2:

Point	Distance to $\left(\frac{5}{3}, 4\right)$	Distance to $\left(5, \frac{4}{3}\right)$	Cluster assigned
(1, 5)	$\sqrt{1.4444}$	$\sqrt{29.4444}$	1
(2, 4)	$\sqrt{0.1111}$	$\sqrt{16.1111}$	1
(4, 1)	$\sqrt{14.4444}$	$\sqrt{1.1111}$	2
(5, 2)	$\sqrt{15.1111}$	$\sqrt{0.4444}$	2
(2, 3)	$\sqrt{1.1111}$	$\sqrt{11.7778}$	1
(6, 1)	$\sqrt{27.7778}$	$\sqrt{1.1111}$	2

Updating the centroids:

$$\mu_1 = \frac{(1,5)+(2,4)+(2,3)}{3} = \left(\frac{5}{3}, 4\right)$$

$$\mu_2 = \frac{(4,1)+(5,2)+(6,1)}{3} = \left(5, \frac{4}{3}\right)$$

As there is no change in the centroids, the algorithm terminates.